

第五章

电磁波的辐射

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5.1 电磁场的矢势和标势

5.2 推迟势

5.3 电偶极辐射

5.4 半波天线和天线阵

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第五章

5.1 电磁场的矢势和标势

- 用势描述电磁场
- 规范变换和规范不变性
- 达朗贝尔方程

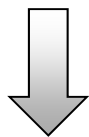
静电场、静磁场的势

➤ 静电场基本方程

$$\nabla \times \vec{E} = 0$$

$$\nabla \cdot \vec{D} = \rho$$

无旋场（保守力场）



引入标量势 φ

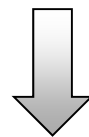
$$\vec{E} = -\nabla \varphi$$

➤ 静磁场基本方程

$$\nabla \times \vec{H} = \vec{J}$$

$$\nabla \cdot \vec{B} = 0$$

有旋场、无散场（无源场）



引入矢量势 $\vec{A}$

$$\vec{B} = \nabla \times \vec{A}$$

矢势A与B的关系仍成立

- Maxwell方程组

$$\begin{aligned}\nabla \times \vec{E} &= -\frac{\partial \vec{B}}{\partial t} \\ \nabla \times \vec{H} &= \vec{J} + \frac{\partial \vec{D}}{\partial t} \\ \nabla \cdot \vec{D} &= \rho \\ \nabla \cdot \vec{B} &= 0\end{aligned}$$

$$\vec{D} = \varepsilon_0 \vec{E} \quad \vec{B} = \mu_0 \vec{H}$$

$$\nabla \cdot \vec{B} = 0$$



引入矢量势A

$$\vec{B} = \nabla \times \vec{A}$$

由B的无源性引入A，在一般情况下B与A的关系仍然成立

非静电场情况下标势不适用

- Maxwell方程组

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\nabla \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$$

$$\nabla \cdot \vec{D} = \rho$$

$$\nabla \cdot \vec{B} = 0$$

$$\vec{D} = \epsilon_0 \vec{E} \quad \vec{B} = \mu_0 \vec{H}$$

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

电场不再是无旋场



引入标量势 ϕ

$$\vec{E} = -\nabla \phi$$

电场 E 不再是无旋场，
不能单独用标势 ϕ 描述

电磁场的势

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \quad \text{电场} E \text{受到磁场的激发, 有旋场}$$

$$\nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0} \quad \text{电场} E \text{受到电荷的激发, 有源场}$$

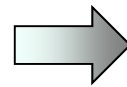
➤ 电磁场的势（矢势+标势）

Maxwell方程组第一式

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

矢量势 $\vec{A}$

$$\vec{B} = \nabla \times \vec{A}$$



$$\nabla \times \left(\vec{E} + \frac{\partial \vec{A}}{\partial t} \right) = 0$$

无旋场

电磁场的势

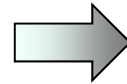
➤ 电磁场的势（矢势+标势）

Maxwell方程组第一式

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

矢量势 $\vec{A}$

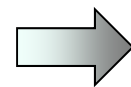
$$\vec{B} = \nabla \times \vec{A}$$



$$\nabla \times \left(\vec{E} + \frac{\partial \vec{A}}{\partial t} \right) = 0$$

无旋场

$$\Rightarrow \vec{E} + \frac{\partial \vec{A}}{\partial t} = -\nabla \varphi$$



$$\vec{E} = -\nabla \varphi - \frac{\partial \vec{A}}{\partial t}$$

电磁场的势

➤ 电磁场的势（矢势+标势）

$$\left\{ \begin{array}{l} \vec{E} = -\nabla \varphi - \frac{\partial \vec{A}}{\partial t} \\ \vec{B} = \nabla \times \vec{A} \end{array} \right. \quad \begin{array}{l} \varphi \text{失去了静电场中势能的概念,} \\ \text{高频系统中电压的概念也失去意义} \end{array}$$

电磁场中的磁场和电场是相互作用的整体，
必须把矢势和标势作为整体来描述电磁场！

引入矢势和标势是因为有时候直接解电磁场方程比较困难

规范变换

➤ 电磁场的势

$$\left\{ \begin{array}{l} \vec{E} = -\nabla \phi - \frac{\partial \vec{A}}{\partial t} \\ \vec{B} = \nabla \times \vec{A} \end{array} \right.$$

给定 E 和 B ，不对应唯一的 A 和 ϕ

设 ϕ 为任一时空函数

$$\text{令} \left\{ \begin{array}{l} \vec{A} \rightarrow \vec{A}' = \vec{A} + \nabla \phi \\ \phi \rightarrow \phi' = \phi - \frac{\partial \phi}{\partial t} \end{array} \right. \Rightarrow \left\{ \begin{array}{l} \nabla \times \vec{A}' = \nabla \times \vec{A} + \nabla \times \nabla \phi = \vec{B} \\ -\nabla \phi' - \frac{\partial \vec{A}'}{\partial t} = -\nabla \phi - \frac{\partial \vec{A}}{\partial t} = \vec{E} \end{array} \right.$$

规范变换

➤ 规范变换

$$\left\{ \begin{array}{l} \vec{A} \rightarrow \vec{A}' = \vec{A} + \nabla \phi \\ \phi \rightarrow \phi' = \phi - \frac{\partial \phi}{\partial t} \end{array} \right. \quad \text{每一组 } (\vec{A}, \phi) \text{ 称为一种规范}$$

$(\vec{A}, \phi)$ 和 $(\vec{A}', \phi')$ 描述同一电磁场，上式称为规范变换

规范不变性

用势来描述电磁场，客观规律和所选择的规范无关，
物理量和物理规律保持不变

势的规范

➤ 电磁场的势

$$\left\{ \begin{array}{l} \vec{E} = -\nabla\varphi - \frac{\partial\vec{A}}{\partial t} \\ \vec{B} = \nabla \times \vec{A} \end{array} \right.$$

从数学上看，这里只给出了 A 的旋度，并没有给出 A 的散度

A 的散度可以取任意值，每一种选择对应一种规范

(1) 库伦 (Coulomb) 规范

(2) 洛伦兹 (Lorentz) 规范

势的规范

(1) 库伦 (Coulomb) 规范

辅助条件 $\nabla \cdot \vec{A} = 0$

$$\vec{E} = -\nabla \varphi - \frac{\partial \vec{A}}{\partial t}$$

E 的纵向场由 φ 描述, $-\nabla \varphi$ 对应库仑场
 E 的横向场由 A 描述, $-\frac{\partial \vec{A}}{\partial t}$ 对应感应场

(2) 洛伦兹 (Lorentz) 规范

辅助条件 $\nabla \cdot \vec{A} + \frac{1}{c^2} \frac{\partial \varphi}{\partial t} = 0$

势的基本方程化为特别简单的对称形式, 在基本理论研究和解决实际辐射问题中特别方便

势满足的基本方程

➤ 电磁场的势

$$\left\{ \begin{array}{l} \vec{E} = -\nabla \varphi - \frac{\partial \vec{A}}{\partial t} \\ \vec{B} = \nabla \times \vec{A} \end{array} \right.$$

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\nabla \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$$

$$\nabla \cdot \vec{D} = \rho$$

$$\nabla \cdot \vec{B} = 0$$

$$\vec{D} = \varepsilon_0 \vec{E} \quad \vec{B} = \mu_0 \vec{H}$$

势满足的基本方程

$$\left\{ \begin{array}{l} \underline{\vec{E} = -\nabla \varphi - \frac{\partial \vec{A}}{\partial t}} \\ \underline{\vec{B} = \nabla \times \vec{A}} \end{array} \right.$$

$$\nabla \times \vec{H} = \vec{J} + \varepsilon_0 \frac{\partial \vec{E}}{\partial t}$$

$$\nabla \cdot \vec{D} = \rho$$

$$\nabla \times \underline{\vec{B}} = \mu_0 \vec{J} + \underline{\mu_0 \varepsilon_0 \frac{\partial \vec{E}}{\partial t}}$$

$$\Rightarrow \nabla \times (\underline{\nabla \times \vec{A}}) = \mu_0 \vec{J} + \mu_0 \varepsilon_0 \frac{\partial}{\partial t} \left(\underline{-\nabla \varphi - \frac{\partial \vec{A}}{\partial t}} \right)$$

$$\Rightarrow \nabla \times (\nabla \times \vec{A}) = \mu_0 \vec{J} - \mu_0 \varepsilon_0 \frac{\partial}{\partial t} \nabla \varphi - \mu_0 \varepsilon_0 \frac{\partial^2 \vec{A}}{\partial t^2}$$

势满足的基本方程

$$\Rightarrow \underbrace{\nabla \times (\nabla \times \vec{A})}_{\nabla(\nabla \cdot \vec{A}) - \nabla^2 \vec{A}} = \mu_0 \vec{J} - \underbrace{\mu_0 \varepsilon_0}_{\mu_0 \varepsilon_0 = 1/c^2} \frac{\partial}{\partial t} \nabla \varphi - \underbrace{\mu_0 \varepsilon_0}_{\mu_0 \varepsilon_0 = 1/c^2} \frac{\partial^2 \vec{A}}{\partial t^2}$$

$$\Rightarrow \nabla(\nabla \cdot \vec{A}) - \nabla^2 \vec{A} = \mu_0 \vec{J} - \frac{1}{c^2} \frac{\partial}{\partial t} \nabla \varphi - \frac{1}{c^2} \frac{\partial^2 \vec{A}}{\partial t^2}$$

$$\Rightarrow \nabla^2 \vec{A} - \frac{1}{c^2} \frac{\partial^2 \vec{A}}{\partial t^2} - \nabla \left(\nabla \cdot \vec{A} + \frac{1}{c^2} \frac{\partial \varphi}{\partial t} \right) = -\mu_0 \vec{J}$$

势满足的基本方程

$$\left\{ \begin{array}{l} \underline{\vec{E} = -\nabla \varphi - \frac{\partial \vec{A}}{\partial t}} \\ \vec{B} = \nabla \times \vec{A} \end{array} \right.$$

$$\nabla \times \vec{H} = \vec{J} + \varepsilon_0 \frac{\partial \vec{E}}{\partial t}$$

$$\nabla \cdot \vec{D} = \rho$$

$$\underline{\nabla \cdot \vec{E}} = \frac{\rho}{\varepsilon_0}$$

$$\Rightarrow \nabla \cdot \left(\underline{-\nabla \varphi - \frac{\partial \vec{A}}{\partial t}} \right) = \frac{\rho}{\varepsilon_0} \Rightarrow \nabla^2 \varphi + \frac{\partial}{\partial t} \nabla \cdot \vec{A} = -\frac{\rho}{\varepsilon_0}$$

势满足的基本方程

电磁场的势

$$\left\{ \begin{array}{l} \vec{E} = -\nabla \varphi - \frac{\partial \vec{A}}{\partial t} \\ \vec{B} = \nabla \times \vec{A} \end{array} \right.$$
$$\nabla \times \vec{H} = \vec{J} + \varepsilon_0 \frac{\partial \vec{E}}{\partial t}$$
$$\nabla \cdot \vec{D} = \rho$$
$$\nabla^2 \vec{A} - \frac{1}{c^2} \frac{\partial^2 \vec{A}}{\partial t^2} - \nabla \left(\nabla \cdot \vec{A} + \frac{1}{c^2} \frac{\partial \varphi}{\partial t} \right) = -\mu_0 \vec{J}$$
$$\nabla^2 \varphi + \frac{\partial}{\partial t} \nabla \cdot \vec{A} = -\frac{\rho}{\varepsilon_0}$$

势满足的基本方程

➤ 势的基本方程

$$\left\{ \begin{array}{l} \nabla^2 \vec{A} - \frac{1}{c^2} \frac{\partial^2 \vec{A}}{\partial t^2} - \nabla \left(\nabla \cdot \vec{A} + \frac{1}{c^2} \frac{\partial \varphi}{\partial t} \right) = -\mu_0 \vec{J} \\ \nabla^2 \varphi + \frac{\partial}{\partial t} \nabla \cdot \vec{A} = -\frac{\rho}{\varepsilon_0} \end{array} \right.$$

采用库伦规范 $\nabla \cdot \vec{A} = 0$

$$\left\{ \begin{array}{l} \nabla^2 \vec{A} - \frac{1}{c^2} \frac{\partial^2 \vec{A}}{\partial t^2} - \frac{1}{c^2} \frac{\partial}{\partial t} \nabla \varphi = -\mu_0 \vec{J} \\ \nabla^2 \varphi = -\frac{\rho}{\varepsilon_0} \end{array} \right.$$

势满足的基本方程

➤ 势的基本方程

$$\left\{ \begin{array}{l} \nabla^2 \vec{A} - \frac{1}{c^2} \frac{\partial^2 \vec{A}}{\partial t^2} - \nabla \left(\nabla \cdot \vec{A} + \frac{1}{c^2} \frac{\partial \varphi}{\partial t} \right) = -\mu_0 \vec{J} \\ \nabla^2 \varphi + \frac{\partial}{\partial t} \nabla \cdot \vec{A} = -\frac{\rho}{\varepsilon_0} \end{array} \right.$$

采用洛伦兹规范 $\nabla \cdot \vec{A} + \frac{1}{c^2} \frac{\partial \varphi}{\partial t} = 0$

$$\left\{ \begin{array}{l} \nabla^2 \vec{A} - \frac{1}{c^2} \frac{\partial^2 \vec{A}}{\partial t^2} = -\mu_0 \vec{J} \\ \nabla^2 \varphi - \frac{1}{c^2} \frac{\partial^2 \varphi}{\partial t^2} = -\frac{\rho}{\varepsilon_0} \end{array} \right.$$

达朗贝尔方程

➤ 达朗贝尔（d'Alembert）方程

采用洛伦兹规范 $\nabla \cdot \vec{A} + \frac{1}{c^2} \frac{\partial \varphi}{\partial t} = 0$

$$\left\{ \begin{array}{ll} \nabla^2 \vec{A} - \frac{1}{c^2} \frac{\partial^2 \vec{A}}{\partial t^2} = -\mu_0 \vec{J} & \text{电流产生矢势波动} \\ \nabla^2 \varphi - \frac{1}{c^2} \frac{\partial^2 \varphi}{\partial t^2} = -\frac{\rho}{\varepsilon_0} & \text{电荷产生标势波动} \end{array} \right.$$

洛伦兹规范下， A 和 φ 具有相同的形式，是非齐次的波动方程，称为达朗贝尔（d'Alembert）方程

以上3个方程是（洛伦兹规范下）电动力学的基本方程组

关于电磁场中引入的矢量势和标量势，描述正确的是

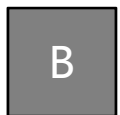
- ☒ A 静磁场和电磁场中都满足 $\vec{B} = \nabla \times \vec{A}$
- ☐ B 用势来描述电磁场，客观规律和所选择的规范有关
- ☐ C φ 仍然具有静电场中势能的概念
- ☒ D 必须把矢势和标势作为整体来描述电磁场

关于达朗贝尔方程，描述正确的是



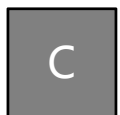
A

相比Maxwell方程组，达朗贝尔方程在基本理论研究和解决实际辐射问题中较为方便



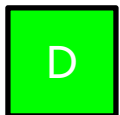
B

电荷产生矢势波动



C

电流产生标势波动



D

达朗贝尔方程是电动力学的基本方程组

平面电磁波的势

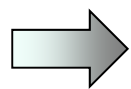
求解：求平面电磁波的势

解：平面电磁波在无电荷、电流的空间中传播

□ 洛伦兹规范下

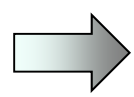
$$\left\{ \begin{array}{l} \nabla^2 \vec{A} - \frac{1}{c^2} \frac{\partial^2 \vec{A}}{\partial t^2} = -\mu_0 \vec{J} \\ \nabla^2 \varphi - \frac{1}{c^2} \frac{\partial^2 \varphi}{\partial t^2} = -\frac{\rho}{\epsilon_0} \end{array} \right. \Rightarrow \left\{ \begin{array}{l} \nabla^2 \vec{A} - \frac{1}{c^2} \frac{\partial^2 \vec{A}}{\partial t^2} = 0 \\ \nabla^2 \varphi - \frac{1}{c^2} \frac{\partial^2 \varphi}{\partial t^2} = 0 \end{array} \right.$$

时谐场



$$\left\{ \begin{array}{l} \vec{A} = \vec{A}_0 e^{i(\vec{k} \cdot \vec{x} - \omega t)} \\ \varphi = \varphi_0 e^{i(\vec{k} \cdot \vec{x} - \omega t)} \end{array} \right. \quad k = \frac{\omega}{c}$$

平面电磁波的势


$$\left\{ \begin{array}{l} \vec{A} = \vec{A}_0 e^{i(\vec{k} \cdot \vec{x} - \omega t)} \\ \varphi = \varphi_0 e^{i(\vec{k} \cdot \vec{x} - \omega t)} \end{array} \right. \quad \Rightarrow \quad \varphi_0 = \frac{c^2}{\omega} \vec{k} \cdot \vec{A}_0$$

洛伦兹规范条件

$$\nabla \cdot \vec{A} + \frac{1}{c^2} \frac{\partial \varphi}{\partial t} = 0$$

给定矢量 $\vec{A}_0$ 即可确定平面电磁波

$$\vec{B} = \nabla \times \vec{A} = i\vec{k} \times \vec{A} = i\vec{k} \times \vec{A}_0 e^{i(\vec{k} \cdot \vec{x} - \omega t)}$$

$$\vec{E} = -\nabla \varphi - \frac{\partial \vec{A}}{\partial t}$$

平面电磁波的势

$$\begin{aligned}
 \vec{E} &= -\nabla \varphi - \frac{\partial \vec{A}}{\partial t} \\
 &= -i\vec{k}\varphi + i\omega\vec{A} \\
 &= -i\vec{k}\left(\frac{c^2}{\omega}\vec{k}\cdot\vec{A}\right) + i\frac{c^2}{\omega}\frac{\omega^2}{c^2}\vec{A}
 \end{aligned}$$

$$\begin{aligned}
 &= -\frac{ic^2}{\omega}\left[\vec{k}\left(\vec{k}\cdot\vec{A}\right) - k^2\vec{A}\right] = -\frac{ic^2}{\omega}\vec{k}\times\left(\vec{k}\times\vec{A}\right) \\
 &= -\frac{c^2}{\omega}\vec{k}\times\vec{B} = -c\vec{e}_k\times\vec{B}
 \end{aligned}$$

$$\vec{A} = \vec{A}_0 e^{i(\vec{k}\cdot\vec{x} - \omega t)}$$

$$\varphi = \varphi_0 e^{i(\vec{k}\cdot\vec{x} - \omega t)}$$

$$\varphi_0 = \frac{c^2}{\omega}\vec{k}\cdot\vec{A}_0$$

$$\vec{B} = \nabla \times \vec{A} = i\vec{k} \times \vec{A}$$

平面电磁波的势

$$\left\{ \begin{array}{l} \vec{B} = \nabla \times \vec{A} = i\vec{k} \times \vec{A} \\ \vec{E} = -\nabla \varphi - \frac{\partial \vec{A}}{\partial t} = -\frac{ic^2}{\omega} \vec{k} \times (\vec{k} \times \vec{A}) \end{array} \right.$$

平面波E和B只与A的横向分量有关，对 A_0 即加上纵向分量，不影响电磁场

若A只有横向分量，即 $\vec{k} \cdot \vec{A} = 0$

$$\Rightarrow \varphi_0 = \frac{c^2}{\omega} \vec{k} \cdot \vec{A}_0 = 0 \quad \Rightarrow \left\{ \begin{array}{l} \vec{B} = \nabla \times \vec{A} = i\vec{k} \times \vec{A} \\ \vec{E} = -\nabla \varphi - \frac{\partial \vec{A}}{\partial t} = i\omega \vec{A} \\ (\vec{k} \cdot \vec{A} = 0) \end{array} \right.$$

平面电磁波的势

□ 在库伦规范下求解

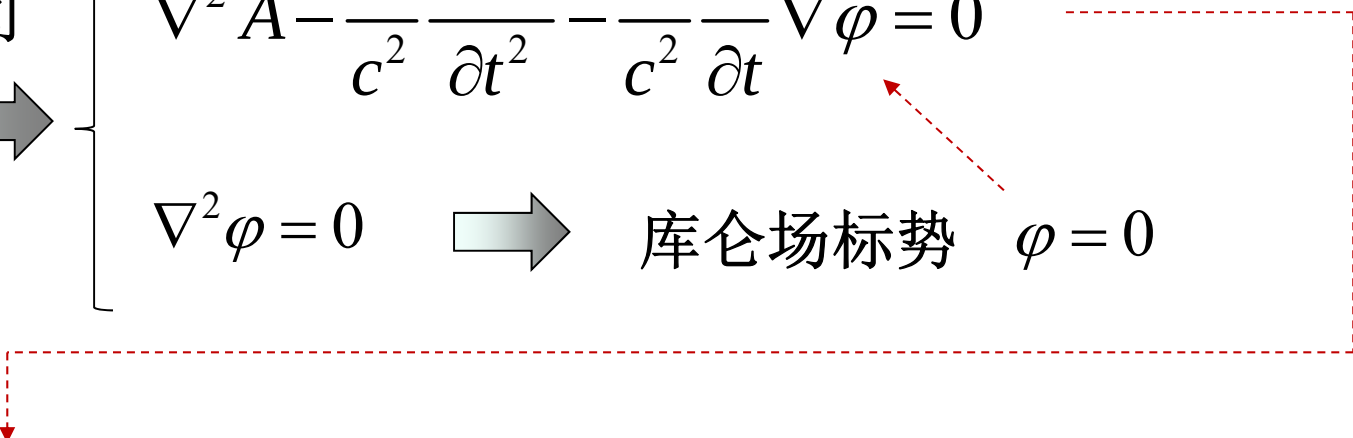
$$\left\{ \begin{array}{l} \nabla^2 \vec{A} - \frac{1}{c^2} \frac{\partial^2 \vec{A}}{\partial t^2} - \frac{1}{c^2} \frac{\partial}{\partial t} \nabla \varphi = -\mu_0 \vec{J} \\ \nabla^2 \varphi = -\frac{\rho}{\varepsilon_0} \\ \text{库伦规范} \quad \nabla \cdot \vec{A} = 0 \end{array} \right.$$

自由空间


$$\left\{ \begin{array}{l} \nabla^2 \vec{A} - \frac{1}{c^2} \frac{\partial^2 \vec{A}}{\partial t^2} - \frac{1}{c^2} \frac{\partial}{\partial t} \nabla \varphi = 0 \\ \nabla^2 \varphi = 0 \end{array} \right. \quad \Rightarrow \quad \begin{array}{l} \text{全空间无电荷情况下} \\ \text{库仑场标势} \quad \varphi = 0 \end{array}$$

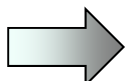
平面电磁波的势

自由空间 

$$\left\{ \begin{array}{l} \nabla^2 \vec{A} - \frac{1}{c^2} \frac{\partial^2 \vec{A}}{\partial t^2} - \frac{1}{c^2} \frac{\partial}{\partial t} \nabla \varphi = 0 \\ \nabla^2 \varphi = 0 \end{array} \right. \Rightarrow \text{库仑场标势 } \varphi = 0$$


$$\nabla^2 \vec{A} - \frac{1}{c^2} \frac{\partial^2 \vec{A}}{\partial t^2} = 0 \Rightarrow \vec{A} = \vec{A}_0 e^{i(\vec{k} \cdot \vec{x} - \omega t)}$$

库伦规范 $\nabla \cdot \vec{A} = 0$ ($\vec{k} \cdot \vec{A} = 0$) 确保 A_0 只有横向分量

 $\vec{B} = i\vec{k} \times \vec{A} \quad \vec{E} = i\omega \vec{A} \quad (\vec{k} \cdot \vec{A} = 0)$

平面电磁波的势

➤ 库伦 (Coulomb) 规范 $\nabla \cdot \vec{A} = 0$

- 标势 ϕ 描述库伦作用, 由电荷 ρ 分布求出
- E 的纵向场由 ϕ 描述, $-\nabla \phi$ 对应库仑场
- 矢势 A 只有横向分量, 描述电磁波的两偏振方向

➤ 洛伦兹 (Lorentz) 规范 $\nabla \cdot \vec{A} + \frac{1}{c^2} \frac{\partial \phi}{\partial t} = 0$

势的基本方程化为特别简单的对称形式, 在基本理论研究和解决实际辐射问题中特别方便

关于势的规范，描述正确的是

- ☒ A 库伦规范条件下，矢势A只有横向分量，描述电磁波的两偏振方向
- ☒ B 库伦规范条件下，标势 φ 描述库伦作用，由电荷 ρ 分布求出
- ☐ C 库伦规范条件为 $\nabla \cdot \vec{A} + \frac{1}{c^2} \frac{\partial \varphi}{\partial t} = 0$
- ☒ D 库伦规范条件下，E的纵向场由 φ 描述， $-\nabla \varphi$ 对应库仑场
- ☒ E 解决实际辐射问题中，洛伦兹规范较为方便

第五章

5.1 电磁场的矢势和标势

5.2 推迟势

达朗贝尔方程

➤ 达朗贝尔（d'Alembert）方程

采用洛伦兹规范 $\nabla \cdot \vec{A} + \frac{1}{c^2} \frac{\partial \varphi}{\partial t} = 0$

$$\left\{ \begin{array}{ll} \nabla^2 \vec{A} - \frac{1}{c^2} \frac{\partial^2 \vec{A}}{\partial t^2} = -\mu_0 \vec{J} & \text{电流产生矢势波动} \\ \nabla^2 \varphi - \frac{1}{c^2} \frac{\partial^2 \varphi}{\partial t^2} = -\frac{\rho}{\varepsilon_0} & \text{电荷产生标势波动} \end{array} \right.$$

洛伦兹规范下， A 和 φ 具有相同的形式，是非齐次的波动方程，称为达朗贝尔（d'Alembert）方程

以上3个方程是（洛伦兹规范下）电动力学的基本方程组

推迟势

达朗贝尔（d'Alembert）方程

$$\left\{ \begin{array}{l} \nabla^2 \varphi - \frac{1}{c^2} \frac{\partial^2 \varphi}{\partial t^2} = -\frac{\rho}{\varepsilon_0} \\ \nabla^2 \vec{A} - \frac{1}{c^2} \frac{\partial^2 \vec{A}}{\partial t^2} = -\mu_0 \vec{J} \end{array} \right.$$

洛伦兹规范

$$\nabla \cdot \vec{A} + \frac{1}{c^2} \frac{\partial \varphi}{\partial t} = 0$$

达朗贝尔
方程的解为



$$\left\{ \begin{array}{l} \varphi(\vec{x}, t) = \frac{1}{4\pi\varepsilon_0} \int_V \frac{\rho(\vec{x}', tr)}{r} dV' \\ \vec{A}(\vec{x}, t) = \frac{\mu_0}{4\pi} \int_V \frac{\vec{J}(\vec{x}', tr)}{r} dV' \end{array} \right. \quad \begin{array}{l} r = |\vec{x} - \vec{x}'| \\ tr = t - \frac{r}{c} \\ \text{为推迟时间} \end{array}$$

推迟势

证明推迟势是达朗贝尔方程的解（以标势为例）

$$\nabla^2 \varphi - \frac{1}{c^2} \frac{\partial^2 \varphi}{\partial t^2} = -\frac{\rho}{\varepsilon_0} \quad \text{的解为} \quad \varphi(\vec{x}, t) = \frac{1}{4\pi\varepsilon_0} \int_V \frac{\rho(\vec{x}', tr)}{r} dV'$$

证明: $\nabla^2 \varphi = \frac{1}{4\pi\varepsilon_0} \int_V \nabla^2 \left(\frac{\rho}{r} \right) dV'$

$$\nabla^2 \left(\frac{\rho}{r} \right) = \nabla \cdot \nabla \left(\frac{\rho}{r} \right) = \nabla \cdot \left(\frac{1}{r} \nabla \rho + \rho \nabla \frac{1}{r} \right)$$

$$\boxed{\nabla \cdot (\varphi \vec{f}) = \nabla \varphi \cdot \vec{f} + \varphi \nabla \cdot \vec{f}} \quad \rightarrow \quad = \frac{1}{r} \nabla^2 \rho + 2 \nabla \rho \cdot \nabla \frac{1}{r} + \rho \nabla^2 \frac{1}{r}$$

推迟势

$$\nabla^2 \left(\frac{\rho}{r} \right) = \nabla \cdot \nabla \left(\frac{\rho}{r} \right) = \frac{1}{r} \nabla^2 \rho + 2 \nabla \rho \cdot \nabla \frac{1}{r} + \rho \nabla^2 \frac{1}{r}$$

$$\nabla \rho = \nabla \rho(\vec{x}', tr) = \frac{\partial \rho}{\partial tr} \nabla tr = \frac{\partial \rho}{\partial t} \nabla tr = -\frac{1}{c} \dot{\rho} \nabla r = -\frac{1}{c} \dot{\rho} \frac{\vec{r}}{r}$$

对 $\vec{x}$ 微分, 而非 $\vec{x}'$

$$tr = t - \frac{r}{c} = t - \frac{|\vec{x} - \vec{x}'|}{c}$$

$$\dot{\rho} \equiv \frac{\partial \rho}{\partial t}$$

$$\nabla^2 \rho = \nabla \cdot \nabla \rho = \nabla \cdot \left(-\frac{1}{c} \dot{\rho} \frac{\vec{r}}{r} \right) = -\frac{1}{c} \frac{\vec{r}}{r} \cdot \nabla \dot{\rho} - \frac{\dot{\rho}}{c} \nabla \cdot \left(\frac{\vec{r}}{r} \right)$$

$$= -\frac{1}{c} \frac{\vec{r}}{r} \cdot \left(-\frac{1}{c} \ddot{\rho} \frac{\vec{r}}{r} \right) - \frac{\dot{\rho}}{c} \frac{2}{r} = \frac{1}{c^2} \ddot{\rho} - \frac{2\dot{\rho}}{cr}$$

推迟势

$$\nabla^2 \left(\frac{\rho}{r} \right) = \nabla \cdot \nabla \left(\frac{\rho}{r} \right) = \frac{1}{r} \nabla^2 \rho + 2 \nabla \rho \cdot \nabla \frac{1}{r} + \rho \nabla^2 \frac{1}{r}$$

$$\nabla^2 \rho = \frac{1}{c^2} \ddot{\rho} - \frac{2\dot{\rho}}{cr}$$

$$\nabla \rho = -\frac{1}{c} \dot{\rho} \frac{\vec{r}}{r}$$

$$\Rightarrow \nabla^2 \left(\frac{\rho}{r} \right) = \frac{1}{c^2 r} \ddot{\rho} - \frac{2\dot{\rho}}{cr^2} + 2 \left(-\frac{\dot{\rho} \vec{r}}{cr} \right) \cdot \left(-\frac{\vec{r}}{r^3} \right) + \rho \nabla^2 \frac{1}{r}$$

$$\Rightarrow \nabla^2 \left(\frac{\rho}{r} \right) = \frac{1}{c^2 r} \ddot{\rho} - 4\pi \rho \delta(\vec{x} - \vec{x}')$$

$$\nabla^2 \frac{1}{r} = -4\pi \delta(\vec{x} - \vec{x}')$$

推迟势

$$\nabla^2 \varphi = \frac{1}{4\pi\epsilon_0} \int_V \nabla^2 \left(\frac{\rho}{r} \right) dV'$$

$$\nabla^2 \left(\frac{\rho}{r} \right) = \frac{1}{c^2 r} \ddot{\rho} - 4\pi\rho\delta(\vec{x} - \vec{x}')$$

$$\Rightarrow \nabla^2 \varphi = \frac{1}{4\pi\epsilon_0} \int_V \left[\frac{1}{c^2 r} \ddot{\rho} - 4\pi\rho\delta(\vec{x} - \vec{x}') \right] dV'$$

$$= \frac{1}{c^2} \frac{1}{4\pi\epsilon_0} \int_V \frac{1}{r} \ddot{\rho} dV' - \frac{1}{\epsilon_0} \int_V \rho\delta(\vec{x} - \vec{x}') dV'$$

$$= \frac{1}{c^2} \frac{\partial^2 \varphi}{\partial t^2} - \frac{\rho}{\epsilon_0}$$

$$\varphi(\vec{x}, t) = \frac{1}{4\pi\epsilon_0} \int_V \frac{\rho(\vec{x}', tr)}{r} dV'$$

推迟势

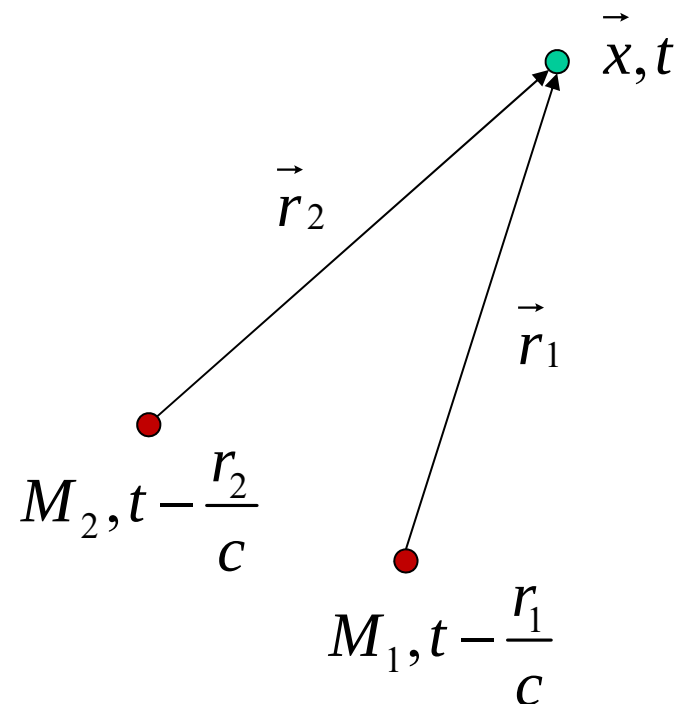
⇒
$$\nabla^2 \varphi - \frac{1}{c^2} \frac{\partial^2 \varphi}{\partial t^2} = -\frac{\rho}{\varepsilon_0} \quad \text{得证}$$

同样的方法，可以证明矢势 A 是达朗贝尔方程的解

推迟势

➤ 推迟势

$$\left\{ \begin{array}{l} \varphi(\vec{x}, t) = \frac{1}{4\pi\epsilon_0} \int_V \frac{\rho(\vec{x}', tr)}{r} dV' \\ \vec{A}(\vec{x}, t) = \frac{\mu_0}{4\pi} \int_V \frac{\vec{J}(\vec{x}', tr)}{r} dV' \end{array} \right.$$



空间 $\mathbf{x}$ 点在 t 时刻的势，
由电荷、电流在较早时刻 ($t-r/c$) 贡献的

空间 $\mathbf{x}$ 点在 t 时刻的电磁场，
是电荷、电流在不同时刻激发的

推迟势

➤ 推迟势（用方括号表示时间推迟）

$$\left\{ \begin{array}{l} \varphi(\vec{x}, t) = \frac{1}{4\pi\epsilon_0} \int_V \frac{\rho(\vec{x}', tr)}{r} dV' = \frac{1}{4\pi\epsilon_0} \int_V \frac{[\rho(\vec{x}')] }{r} dV' \\ \vec{A}(\vec{x}, t) = \frac{\mu_0}{4\pi} \int_V \frac{\vec{J}(\vec{x}', tr)}{r} dV' = \frac{\mu_0}{4\pi} \int_V \frac{[\vec{J}(\vec{x}')] }{r} dV' \end{array} \right.$$

杰斐逊(Jefimenko)方程

➤ 杰斐逊(Jefimenko)方程

$$\vec{E} = -\nabla\varphi - \frac{\partial\vec{A}}{\partial t} = \frac{1}{4\pi\epsilon_0} \int_V \left(\frac{[\rho]}{r^2} \hat{r} + \frac{[\dot{\rho}]}{cr} \hat{r} - \frac{[\vec{J}]}{c^2 r} \right) dV'$$

$$\vec{B} = \nabla \times \vec{A} = \frac{\mu_0}{4\pi} \int_V \left(\frac{[\vec{J}]}{r^2} + \frac{[\dot{\vec{J}}]}{cr} \right) \times \hat{r} dV'$$

推迟势

$$\varphi(\vec{x}, t) = \frac{1}{4\pi\epsilon_0} \int_V \frac{[\rho(\vec{x}')] }{r} dV', \quad \vec{A}(\vec{x}, t) = \frac{\mu_0}{4\pi} \int_V \frac{[\vec{J}(\vec{x}')] }{r} dV'$$

交变电流的推迟势

➤ 交变电流的推迟势

推迟势

$$\vec{A}(\vec{x}, t) = \frac{\mu_0}{4\pi} \int_V \frac{\vec{J}\left(\vec{x}', t - \frac{r}{c}\right)}{r} dV'$$

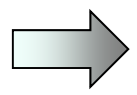
假设电流 $\vec{J}$ 和电荷 ρ 按照一定频率变化

$$\vec{J}(\vec{x}', t) = \vec{J}(\vec{x}') e^{-i\omega t} \quad \rho(\vec{x}, t) = \rho(\vec{x}) e^{-i\omega t}$$

➡

$$\vec{A}(\vec{x}, t) = \frac{\mu_0}{4\pi} \int_V \frac{\vec{J}(\vec{x}')}{r} e^{-i\omega\left(t - \frac{r}{c}\right)} dV'$$

交变电流的推迟势


$$\vec{A}(\vec{x}, t) = \frac{\mu_0}{4\pi} \int_V \frac{\vec{J}(\vec{x}')}{r} e^{-i\omega\left(t - \frac{r}{c}\right)} dV'$$

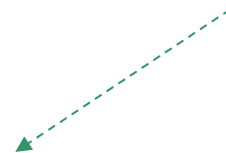
$$= \frac{\mu_0}{4\pi} \int_V \frac{\vec{J}(\vec{x}')}{r} e^{ikr - i\omega t} dV'$$

$$= \underline{\vec{A}(\vec{x})} e^{-i\omega t}$$

$k = \frac{\omega}{c}$ 是波数

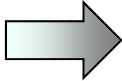
e^{ikr} 是推迟因子

电磁波传至场点时的相位滞后


$$\vec{A}(\vec{x}) = \frac{\mu_0}{4\pi} \int_V \frac{\vec{J}(\vec{x}')}{r} e^{ikr} dV'$$

交变电流的推迟势

由洛伦兹规范 $\nabla \cdot \vec{A} + \frac{1}{c^2} \frac{\partial \varphi}{\partial t} = 0$

 $\varphi = \frac{c^2}{i\omega} \nabla \cdot \vec{A}$

交变情况下，由 $\vec{J}$ 完全确定电磁场

$$\vec{J} \rightarrow \vec{A}, \varphi \quad \vec{B} = \nabla \times \vec{A} \quad \vec{E} = \frac{ic}{k} \nabla \times \vec{B}$$

关于推迟势，描述正确的是

- ☐ A 空间x点在t时刻的势，
由电荷、电流在同一时刻t贡献的
- ☒ B 空间x点在t时刻的电磁场，是电荷、电流
在不同时刻激发的
- ☒ C 交变情况下，由电流J完全确定矢量势和
标量势
- ☒ D 推迟因子 e^{ikr} 是电磁波传至场点时的相位滞后

第五章

5.1 电磁场的矢势和标势

5.2 推迟势

5.3 矢势展开和电偶极辐射